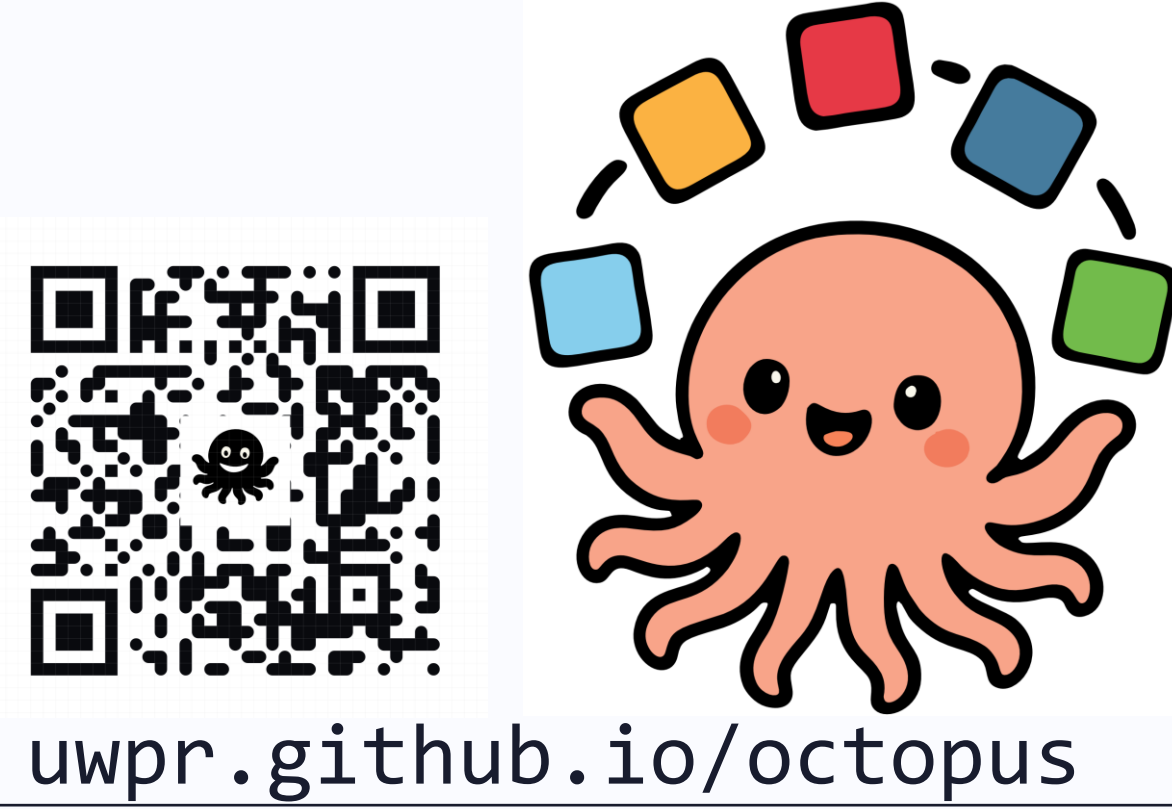


# Octopus: A Web Application for Plate Design with Balanced Block Randomization

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## Overview

**Octopus** is a freely accessible web application for designing balanced, randomized plate layouts for mass spectrometry workflows. It combines proportional allocation of covariate groups across plates with spatial optimization to minimize same-group clustering within plates. It supports repeated-measures designs where samples from the same subject must remain together, and provides real-time visualization, drag-and-drop editing, and integrated quality metrics. All processing happens locally in the browser, requiring no installation and ensuring that data never leaves the user's computer.

## Introduction

Proper experimental design is essential for reducing batch effects and technical variability in large-scale mass spectrometry studies. LC-MS performance drift, differences in sample preparation, and multi-batch processing can introduce confounders that bias downstream interpretation. Randomization is used to mitigate such confounders, but unrestricted randomization can produce imbalanced designs where biological and technical effects become indistinguishable. Block randomization distributes covariates of interest across plates, minimizing such confounding<sup>1</sup>. Proportional distribution ensures balanced representation of covariate groups across batches, while spatial optimization reduces position-dependent clustering within plates. Existing R-based tools (WPM<sup>2</sup>, Omixer<sup>3</sup>) require local installation, limiting accessibility. Web-based alternatives (PlateDesigner<sup>4</sup>, PLAID<sup>5</sup>) address some of these needs but none combine multi-covariate balance, spatial optimization, and repeated-measures constraints in a single interactive tool.

## Octopus Goals

- Balance covariate groups proportionally across plates and rows
- Minimize clustering of same-covariate-group samples within plates
- Keep repeated-measures samples together: same row, or same plate
- Spread QC samples evenly across plates and rows
- Support interactive visualization and post-randomization editing
- Run in the browser; no install, data stays local

## Methods

**Input:** A CSV metadata file, one row per sample. User selects a Sample ID column and one or more covariate columns

Covariate Key = concatenation of selected columns (e.g. Treatment | Dose | Time)

Optional: QC/reference column, Subject ID column with a same-row or same-plate constraint, and plate dimensions (default 8 × 12)

### Output:

- Interactive plate visualization with covariate-colored cells
- CSV: original metadata augmented with plate number and well position
- *Excel*: color-coded plate layouts with sample IDs and covariate values
- *Injection sequence CSV* (Thermo format): plate-ordered worklist with configurable file naming template, system suitability injections

**Quality Scores:** Two scores (0–100) quantify how well a layout meets its design goals

**Balance** Weighted deviation of per-covariate-group proportions from their globally expected values, per plate.  
100 = perfectly balanced

**Clustering**  $(1 - C / C_{\max}) \times 100$   
C is the number of same-group adjacencies  
 $C_{\max}$  is the total possible adjacencies  
100 = no same-group neighbors

## Results

### Case 1 – Multi-plate balanced covariate distribution

936 samples | 10 plates | 37 covariate groups

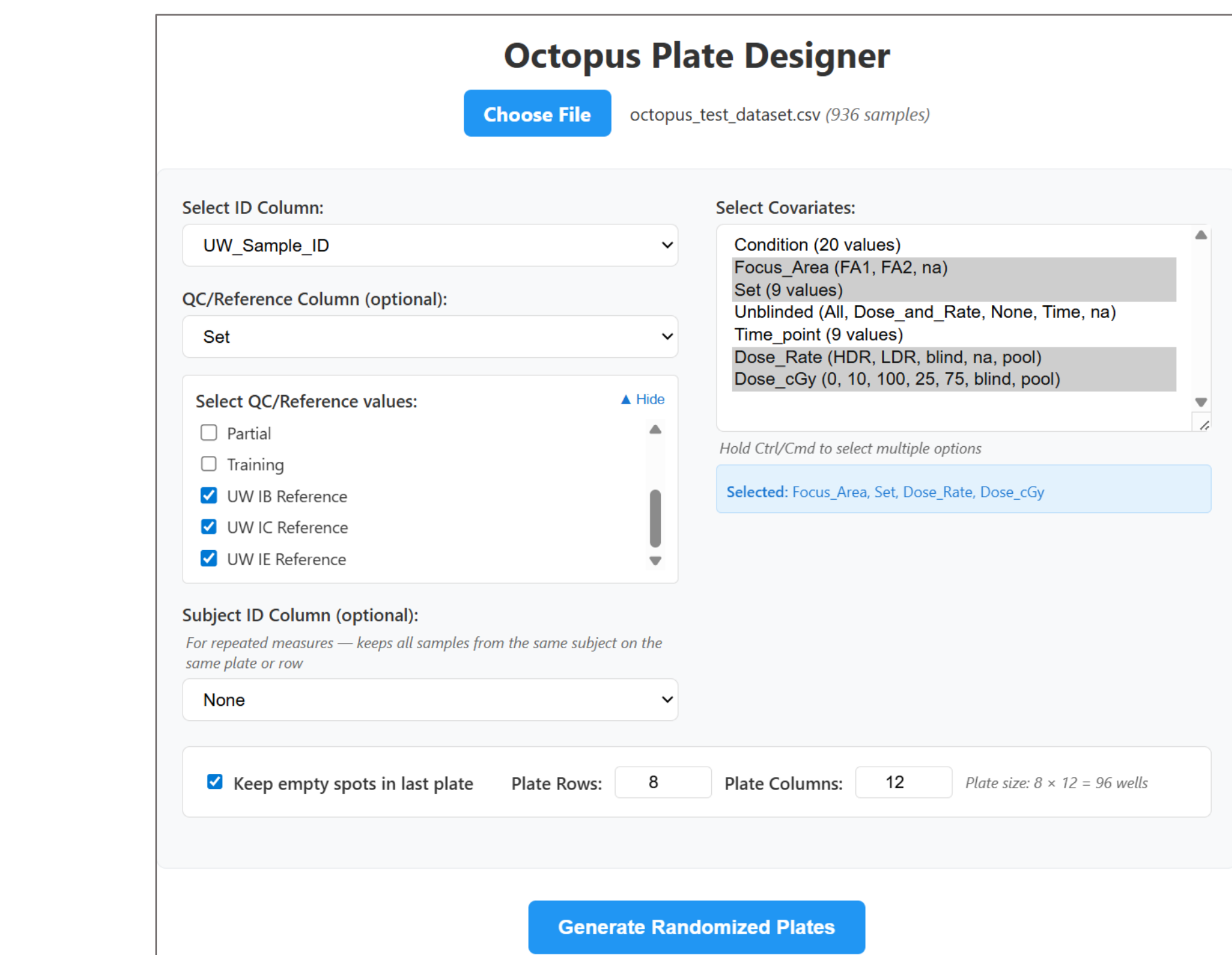


Figure 1. Configuration panel guides the user from CSV import to plate generation.

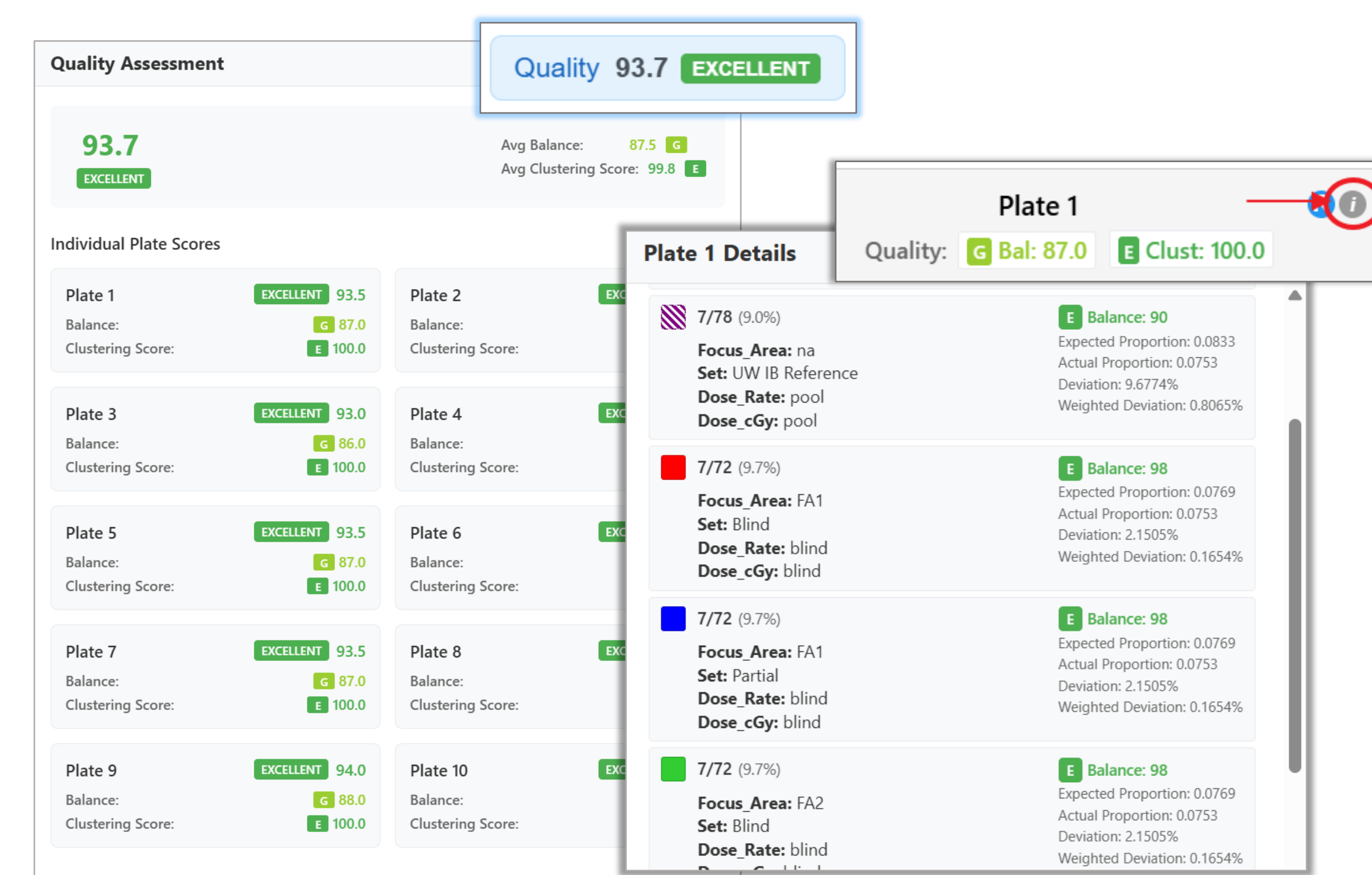


Figure 3. Summary of quality scores per plate (left panel). Covariate group distribution numbers and balance scores can be viewed for each plate by clicking the 'i' icon in the top right corner of the plate header (right panel).

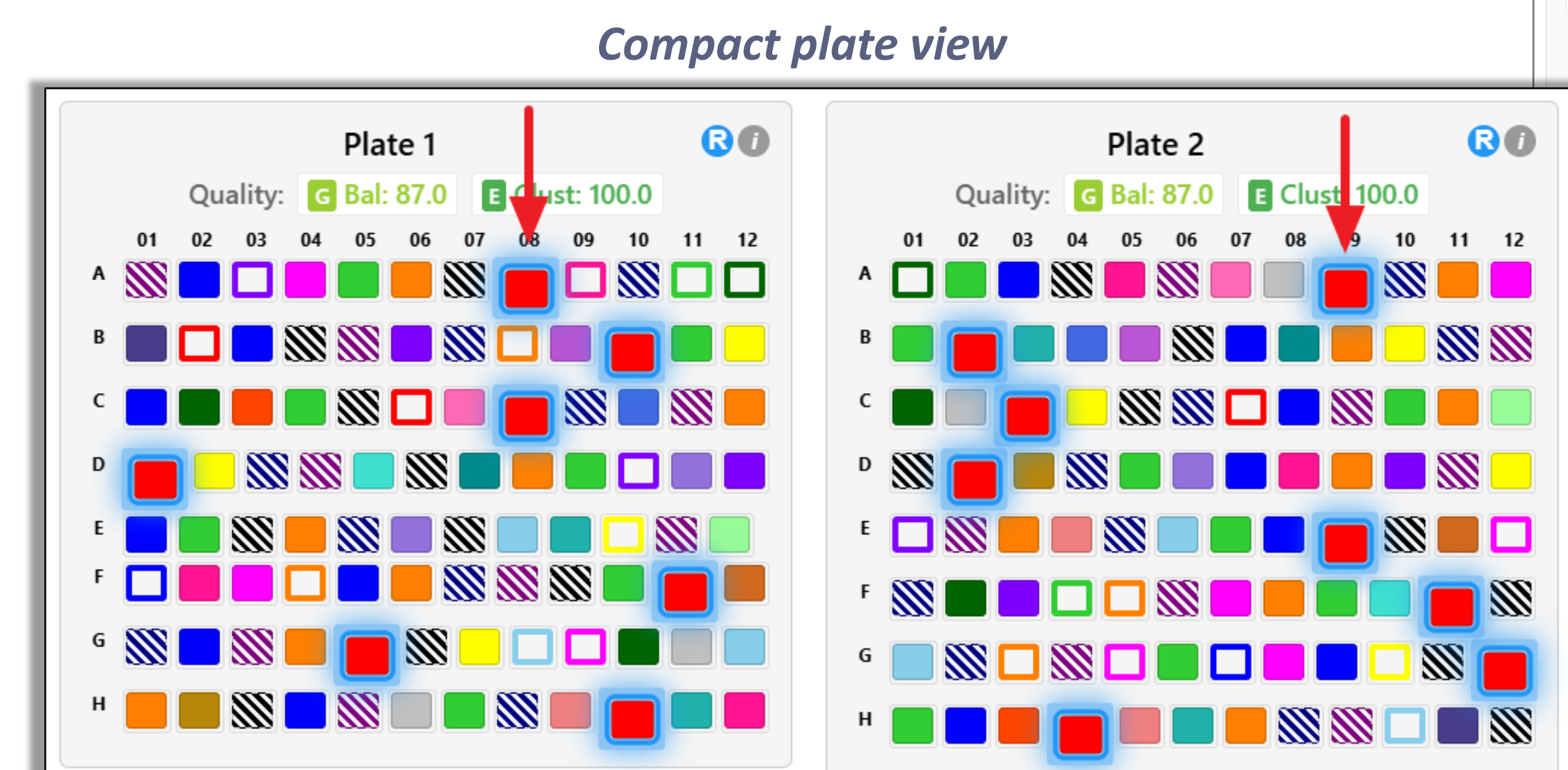


Figure 2a. 936 samples are distributed across 10 plates. 2 of 10 plates shown. The highlighted wells are samples belonging to the covariate group selected in the Covariate Summary View (Figure 2b).



Figure 4. Drag-and-drop editing with live quality score update. **Top:** A user moves a sample (indicated by arrows) from C:12 to C:5 to reduce same-covariate-group adjacencies. **Bottom:** The Clustering score updates immediately from 94.0 to 96.0, while Balance is unchanged.

### Case 2 – Repeated measures

96 samples | 16 QC | 29 patients x 2-5 timepoints with Same Row constraint

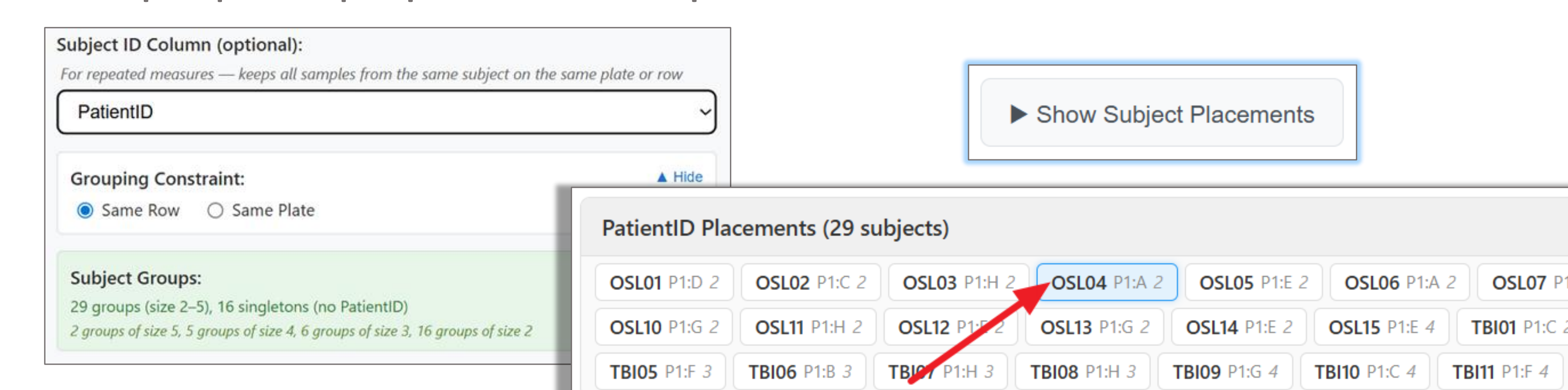


Figure 6. Samples from the same patients are placed on the same rows. Per-subject plate and row assignments can be viewed in the **Patient ID Placements** panel. Highlighting one patient on the panel highlights all the wells where samples from the patient were placed.

## Conclusions

To our knowledge, Octopus is the only browser-based tool that combines multi-covariate balance, QC sample distribution, and repeated-measures subject grouping in a single interactive application. It supports drag-and-drop editing, per-plate re-randomization, and covariate and subject highlighting for easy design verification. Layouts can be exported as color-coded Excel files for producing print-ready worksheets for benchtop use. It is free, open-source, and requires no installation. Since all processing happens locally in the browser, no data is transmitted externally, making it suitable for use with clinical metadata.

## References

1. Burger et al. *Journal of Proteome Research* 2021
2. Borges et al. *Bioinformatics* 2021
3. Sinke et al. *Bioinformatics* 2021
4. Suprun et al. *Bioinformatics* 2019
5. Francisco Rodriguez et al. *Artificial Intelligence in the Life Sciences* 2023

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